



HABIB UNIVERSITY
Digital Logic and Design
EE-172 / CS-130 / CE-222
Fall 2025 (L2)

Quiz # 1

Student ID: _____

Student Name: _____

Date: September 18, 2025

Time: 25 minutes

Max points: 24

Instructions:

- i. **Smart watches, laptops, and similar electronics are strictly NOT allowed.**
- ii. **Answer sheets should contain all steps, working, explanations, and assumptions.**
- iii. Attempt the quiz in the spaces provided for each question.
- iv. You are not allowed to ask/share your method or answer with your peers. The work submitted by you is solely your own work. Any violation of this will be the violation of HU Honor code and proper action will be taken as per university policy if found to be involved in such an activity.

No.	Questions	Pts
1	Write the decimal equivalent of 1011 if it is interpreted as: a binary number $1x2^3 + 0x2^2 + 1x2^1 + 1x2^0 = 8 + 0 + 2 + 1 = 11)_{10}$	2
	a 2's complement number $1x(-2^3) + 0x2^2 + 1x2^1 + 1x2^0 = -8 + 0 + 2 + 1 = -5)_{10}$ Or Taking 2's complement: - 0101 = -5)_{10}	2
	a Gray code $1011)_G = 1101)_2 = 1x2^3 + 1x2^2 + 0x2^1 + 1x2^0 = 8 + 4 + 0 + 1 = 13)_{10}$	2
	a base-3 number $1011)_3 = 1x3^3 + 0x3^2 + 1x3^1 + 1x3^0 = 27 + 0 + 3 + 1 = 31)_{10}$	2

- 2 Perform the following arithmetic operations in a 5-bit 2's complement system and report if overflow occurs

▪ $10)_{10} - 4)_{10}$

5

$$10)_{10} = 01010)_2$$

$$-4)_{10} = 11100)_2 \text{ [2's complement]}$$

$$\begin{array}{r} 01010)_2 \\ +11100)_2 \\ \hline 100110)_2 \end{array}$$

Since this is subtraction generated extra bit and will be discarded. No overflow occurs in subtraction. Hence the answer is $00110)_2$.

▪ $10)_{10} + 4)_{10}$

5

$$10)_{10} = 01010)_2$$

$$4)_{10} = 00100)_2$$

$$\begin{array}{r} 01010)_2 \\ +00100)_2 \\ \hline 01110)_2 \end{array}$$

Both operands have 0 as the MSB (positive), and the result has the same sign; therefore, no overflow occurs.

- 3 An 8-bit value (7 data bits + 1 parity bit) is transmitted from the processor to main memory. It is received as **10110011**. The system uses even parity. Determine whether the received data contains an error. Briefly justify your answer.

2

Yes, the received data contains an error because the total number of 1s (5) is odd, which violates the even parity rule.

- 4 Which of the following statements are true for the BCD (Binary Coded Decimal) coding system? (Write true or false in front of each option)

3

- Uses more memory than straight binary codes.
- Used in calculators for input and output.
- Is difficult to convert to decimal as compared to binary.
- Equivalent to hex coding system.
- Has 8 invalid codes.
- Performs arithmetic operations less efficiently than straight binary.

- 5 How many bits are required to assign unique binary codes to all 26 English capital letters? Briefly justify your answer.

1

5 bits are required, because $2^5=32$ possible codes can uniquely represent all 26 letters. 6 codes will remain unused here.